Model predictive control strategy for a solar-powered irrigation pump supplying a storage reservoir for a 1-hectare onion field in kenya

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Abstract

The decoupling of stochastic renewable energy generation from critical water demand in off-grid solar irrigation systems presents a fundamental control challenge. This is particularly acute in hydrogeologically constrained environments, where aggressive extraction can lead to irreversible aquifer damage [1]

This study proposes a Hierarchical Model Predictive Control (HMPC) framework that integrates a high-fidelity hydraulic model based on Thiem's equilibrium equations. The architecture utilizes a two-layer strategy: a 6-day High-Level Planner for volumetric allocation and an hourly MPC Regulator for real-time constraint enforcement. Utilizing real-world TMY data for Nairobi, the simulation evaluates system resilience.

Under constrained aquifer conditions, characterized by low transmissivity ($T = 8.7 \times 10^{-5}$ m²/s), a 3000 Wp PV system achieved superior operational results compared to larger systems. While intuition might suggest that more power leads to more water, increasing the PV array size (e.g., to 5000 Wp or more) without a corresponding increase in aquifer yield leads to the Oversizing Paradox.

The larger power input from oversized systems dewaters the borehole too quickly, causing the dynamic drawdown to hit critical safety limits and triggering frequent, long pump lockout periods. Furthermore, larger systems are hindered by a higher start-up power threshold (modeled as 20% of rated capacity), meaning they remain inactive during low-irradiance hours that a 3000 Wp system can successfully utilize.

Under relaxed hydraulic conditions, the "Oversizing Paradox" is mitigated because the aquifer can replenish the borehole nearly as fast as the pump extracts water. This environment is characterized by high transmissivity ($T = 1.0 \times 10^{-1}$ m²/s), which prevents the rapid dewatering of the borehole that typically triggers safety lockouts in constrained geologies. Consequently, a "more is better" relationship between PV capacity and water yield is established, as the system is limited primarily by available solar energy rather than the aquifer's recharge rate.

The results demonstrate that "hydro-aware" control strategies are essential for the sustainable intensification of solar irrigation, confirming that aquifer properties, rather than solar peak power, are the governing constraint in many African contexts.

Introduction

Photovoltaic Water Pumping Systems (PVWPS) are expanding rapidly in the Global South to decouple agricultural productivity from fossil fuels. In these regions, groundwater is a critical buffer against climate variability. However, the transition to solar introduces a mismatch between energy availability and water demand. Conventional design often focuses on Maximum Power Point Tracking (MPPT) to maximize electrical yield, frequently ignoring hydrogeological limits. As noted by Meunier et al. (2023), the governing constraint in many African context is the aquifer's hydraulic properties rather than solar peak power [1]

Sustainable precision irrigation requires a shift toward closed-loop strategies that adapt to environmental dynamics [2]. While MPC has been used for soil moisture regulation, its application to upstream resource protection (borehole hydraulics) remains underrepresented. This research targets that gap, demonstrating how intelligent load shaping can optimize water delivery without compromising hydraulic integrity.

This research addresses these limitations by developing a hydro-aware control paradigm. We propose a Model Predictive Control MPC framework that explicitly integrates groundwater physics into the decision-making process. By shifting the modeling focus from solar-constrained to also aquifer-constrained, we demonstrate how intelligent load shaping can optimize water delivery without compromising hydraulic integrity.

By leveraging the borehole and reservoir as "hydraulic buffers," the MPC framework enables a "batteryless" system architecture. This significantly reduces the Capital Expenditure (CAPEX) and maintenance complexity compared to traditional off-grid systems that rely on expensive electrochemical storage (e.g., Li-Ion batteries) to smooth solar intermittency.

Theoretical Framework and Physical Modeling

The modeling approach adopted in this study is grounded in the aquifer-constrained methodology established by Meunier et al. (2023) [1]

Groundwater Hydrodynamics

The aquifer is modeled using the Thiem equation for steady-state radial flow, assuming the cone of depression stabilizes within hourly intervals. The dynamic drawdown $\left(H_{\{bd\}}\right)$ is defined as:

$$H_{\{bd\}(Q)} = \frac{\left\{ln\left(\frac{r_c}{r_b}\right)\right\}}{\{2\pi T\}} Q \ + \ \beta Q^2$$

Where Q is pumping rate $(m\text{^}3/s)$, T is Transmissivity $(m\text{^}2/s)$, and r_c and r_b are the radii of influence and the borehole.

The Cubic Power-Flow Relation

To link the electrical domain (PV power) to the hydraulic domain (flow rate), we utilize the cubic equilibrium equation. This equation balances the mechanical power supplied by the motor with the hydraulic power required to overcome static lift, dynamic drawdown, and pipe friction:

$$(\beta + \nu L_p)Q^3 + \frac{\left\{\ln\left(\frac{r_c}{r_b}\right)\right\}}{\{(2\pi T)\}}Q^2 + H(b,s)Q - \frac{\{P_{elec} \cdot \eta_{mp}\}}{\{pg\}} = 0$$

The roots of this polynomial determine the feasible flow rate for a given power input. Crucially, if the calculated drawdown exceeds the pump depth (H_{mp}), the flow must be curtailed to zero, creating a hard non-linearity that the controller must predict and avoid.

Model Assumptions and Limitations

To ensure computational tractability within the optimization loop, the following physical and operational assumptions are imposed on the simulation framework:

• **Quasi-Steady-State Flow:** We assume the cone of depression stabilizes within the simulation time step ($\Delta t = 1$ h). This justifies the use of the steady-state Thiem equation.

• **Ideal VSD Operation**: The hydraulic model assumes an ideal Variable Speed Drive (VSD) capable of matching the pump's operating point to the system curve at any flow rate, constrained only by available power. This simplifies the specific Head-Flow ($H - Q$) characteristic curve of the pump.

• **Constant Transmissivity:** The aquifer transmissivity T is modeled as time. This assumes a confined aquifer response where the saturated thickness remains effectively constant despite drawdown.

• **Fixed Radius of Influence:** The radius of the cone of depression (rc) is fixed at a certain value, assuming the pressure disturbance has propagated to a steady boundary condition.

• **Aggregated Efficiency & Losses:**

- The combined motor-pump efficiency (η_{mp}) is fixed at 36%, neglecting efficiency degradation at partial loads.
- PV temperature and cabling losses are aggregated into a static derating factor of 25% ($C_{loss} = 0.25$), ignoring thermal coefficients of voltage.

• **Conservative Storage Dynamics:** The reservoir model ignores evaporative losses, assuming a closed storage system. A "minimum volume buffer of 10% of tank capacity is strictly enforced.

Sizing and Operational Assumptions: The simulation is constrained by the following design parameters, corresponding to a standard small-holder irrigation setup:

- **Farm Scale:** 1 Hectare (10,000 m^2)

- **Reservoir Capacity:** The storage buffer is sized to match one day of peak demand (Vmax = 70 m^3).

- **Pump Starting**: The pump is programmed with a start-up ratio of 0.2 (20% of rated capacity). This means the system physically cannot initiate pumping until the available solar power exceeds this threshold (e.g., 600 W for a 3000 Wp system)

- **Prediction Profiles**: Future solar generation is forecasted using perfect knowledge (in simulation) of the TMY irradiance profile, assuming a "Receding Horizon" where the controller re-optimizes its plan every hour based on the latest tank level and weather data.

- **Crop Tolerance:** The Planner's "Solar Shaping" strategy assumes the onion crop is insensitive to intra-block irrigation timing, provided the cumulative 6-day water requirement is met.

- **Simplified Demand Profile:** The Crop Water Requirement is modeled as a step-wise 6-day volumetric target (300-100-400 m^3), rather than using a dynamic crop coefficient.

Methodology

Hierarchical Control Architecture

The selection of Model Predictive Control (MPC) over classical strategies (e.g., PID or rule-based logic) is driven by the complexity of agricultural dynamics. As detailed in the review by Bwambale et al. (2023), MPC is uniquely suited for precision irrigation due to its ability to handle multivariate (MIMO) processes and, most importantly, to explicitly enforce state and input constraints [2].

To manage the disparate time scales of seasonal water planning and real-time solar fluctuations, a Hierarchical Model Predictive Control (HMPC) structure was selected. This design choice is grounded in the stability frameworks for hierarchical MPC established by Scattolini and Colaneri (2007) [3].

As motivated by their work, this architecture addresses the challenge of systems composed of subsystems operating at different stratifications. The Higher Level (Planner) is tasked with the primary function of the system—in this case, ensuring long-term volumetric water security, while the Lower Level (Regulator) is responsible for handling the fast dynamics and intrinsic physical limitations of the subsystem.

The framework utilizes a multi-rate implementation where the higher layer acts at a slower sampling rate (k) than the lower layer (h). This justifies our choice of a 6-day block for the Planner versus an hourly step for the lower level MPC. This separation allows the Planner to pass a desired control trajectory to the Regulator. Crucially, as defined in the robust design procedure, the lower level must track this target while respecting local robustness constraints. If the physical constraints of the aquifer prevent the realization of the target (e.g., drawdown limit

reached), a negotiation occurs wherein the system implicitly accepts a performance degradation deficit to maintain closed-loop stability and hydraulic integrity

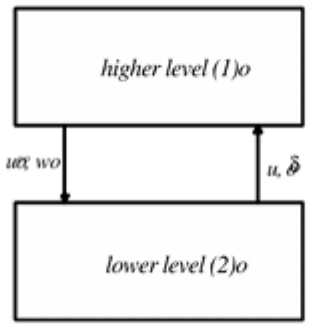

Figure 1: Two level HMPC

The HMPC in this study is implemented in PyCharm IDE (Python) using the scipy optimization suite

Layer 1: High-Level Planner

The Planner operates on a 6-day block horizon (T_{block} = 144h). Its primary objective is to decompose a long-term volumetric target (V_{target}) into daily pumping goals based on forecasted solar availability. Its primary objective is to decompose a long-term volumetric target into daily pumping goals. This aligns with the definition of irrigation scheduling provided by Bwambale et al. (2023), which entails determining the optimal frequency, duration, and quantity of water application [2].

The planner enforces a strict irrigation window between 13:00 and 22:00. This constraint is implemented to align extraction with the specific demand profile of the peri-urban agricultural test site, further stressing the storage capabilities of the system. The allocation logic utilizes a Solar Score (Sd) for each day d in the block. This proportional allocation logic effectively "shapes" the demand profile to match the solar generation curve.

Layer 2: Low Level Regulator

The hourly Regulator solves a constrained optimization problem over a 72-hour prediction horizon. It employs a "Receding Horizon" strategy, where only the first control action is applied before re-optimizing at the next step.

Following the standard MPC formulation described by Bwambale et al. (2023) [2], the Regulator employs a "Receding Horizon" strategy. At each time step k, the optimal control sequence is computed over the prediction horizon N = 72. However, only the first control action u(k|k) is applied to the pump. At the next time step k+1, the horizon shifts forward, and the optimization is re-solved with updated state feedback. This iterative process allows the controller to inherently correct for disturbances (e.g., cloud cover deviations) that were not captured in the initial forecast.

The objective function J is designed to balance three competing goals: minimizing water deficit, preventing aquifer stress, and reducing actuator wear. The weights were tuned empirically based on the simulation requirements:

$$J = \sum_{k=1}^{N} (w_{def}(Vreq - Vpump)^2 + w_{dd}H_{bd}^2 + w_{\Delta u}\Delta u^2)$$

Specific weights used in this implementation are:

- **Deficit Weight (w_{def} = 3.0):** High priority to ensure crop water requirements are met. Different deficit weight values were tested for robust assessment.
- **Drawdown Weight (w_{dd} = 1.0):** Penalizes excessive drawdown to prevent approaching the pump intake limit.
- **Smoothness Weight ($w_{\Delta u}$= 1.0):** Discourages rapid start and stop of the pump.

Implementation

The control strategy is validated against a virtual representation of a 1-hectare onion field located in Nairobi, Kenya. The system architecture consists of a solar-powered submersible pump supplying a storage reservoir, which buffers the immediate irrigation demand. The primary objective of the MPC framework is to determine when and at what capacity the pump should operate. The controller must navigate a critical trade-off: keeping the reservoir sufficiently filled to satisfy the daily crop water requirement of the onion field, while strictly adhering to the volumetric limits of the storage tank and the hydraulic recovery limits of the borehole

The MPC framework was implemented in Python using the `scipy.optimize` suite. The software architecture follows a modular design:

Hydraulic Solver Implementation

The code solves the cubic flow equation using `numpy.roots` to find the physically feasible flow rate for any given power input.

- **Constraint Logic:** If the drawdown hits the motor-pump depth, the solver curtails flow to zero to prevent dry-running.
- **Efficiency:** A fixed motor-pump efficiency of 36% is used.
- **Electrical Derating:** A static derating factor of 25% (C_{loss}=0.25) is applied to account for cabling and thermal losses

Real-Time Irradiance Pipeline

Environmental forcing is generated via the `pvlib` library, fetching Typical Meteorological Year (TMY) data for Nairobi (-1.29, 36.82), covering the range 2005–2016. This irradiance timeline

was generated by extracting the first 4 days of every month from the historical TMY record and concatenating them sequentially (e.g 4 days × 9 months = 36 days).

Cost Function and Tuning Weights

The optimization objective (J) is formulated as a weighted sum of quadratic penalties:

- **Deficit ($w_{def} = 3.0$):** High weight ensuring the delivered volume matches the target provided by the Planner.
- **Drawdown ($w_{dd} = 1.0$):** Protects the aquifer and pump.
- **StartStop= 1.0):** Prevents mechanical stress on the VSD and motor by penalizing rapid pump speed changes.

Simulation Parameters (Relaxed Aquifer)

To establish the upper limit of system performance, the borehole was parameterized to represent "Best Aquifer Conditions," characterized by high transmissivity and low hydraulic resistance.

Table 1: Simulation Parameters

Parameter	Value	Unit	Description
PV Array Size	3000	Wp	Targeted optimal size
Static Water Depth	15.0	m	Level when pump is idle
Transmissivity (T)	1.0×10^{-1}	m^$2/s$	High-yield gravel/karst aquifer
Borehole Radius	0.5	m	Wide-diameter well
Loss Coefficient	10.0	-	Minimal turbulence at intake
Saturated Thickness	100.0	m	Massive vertical column
Motor-Pump Depth	115.0	m	Deep submergence safety
Min. Start Threshold	0.2	-	20% of rated pump power

Simulation parameters (Constrained Aquifer)

Parameter	Value	Unit	Description
Static Water Level	15.0	m	Depth to water when the pump is idle.
Transmissivity	8.7×10^{-5}	m^2/s	Low hydraulic conductivity typical of regional volcanic geology.
Borehole Radius	0.1	m	Standard narrow-diameter well.
Borehole Loss Coefficient	5.0×10^4	s^2 $/m$^5	High turbulent losses at the screen/intake.
Radius of Influence	500	m	Fixed radius for the cone of depression.
Saturated Thickness	62.5	m	The vertical extent of the water-bearing zone.
Motor-Pump Depth	35.0	m	Total depth of the pump intake
Minimum Start Ratio	0.2	-	Required power threshold (20% of rated capacity) to initiate pumping.

Results

In the constrained aquifer scenario, characterized by low transmissivity, the simulation identified a clear performance peak at the **3000 Wp PV array size**. As shown in the deficit analysis, increasing the PV size from 1500 Wp to 3000 Wp significantly reduced the total water deficit. However, increasing the capacity beyond 3000 Wp (e.g., to 5000 Wp) resulted in the **Oversizing Paradox**

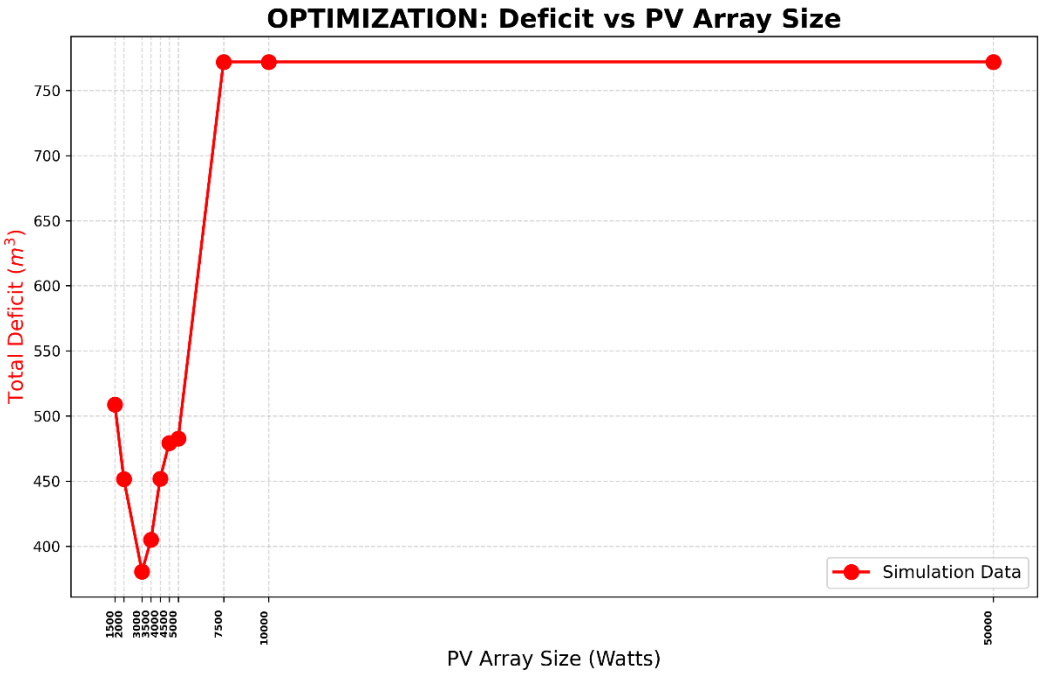

Figure 2: Deficit Analysis under Constrained Aquifer Conditions

Figure 2: The simulation results for a low-transmissivity aquifer illustrate the "Oversizing Paradox." The Total Deficit curve (red) reaches a distinct global minimum at **3000 Wp**, identifying this as the optimal array size for the given hydraulic parameters. Contrary to standard MPPT logic, increasing the PV capacity beyond 3500 Wp results in a sharp *increase* in water deficit. This performance degradation occurs because the oversized arrays dewater the borehole too rapidly, causing the dynamic drawdown to hit the safety limit and triggering frequent, extended pump lockouts.

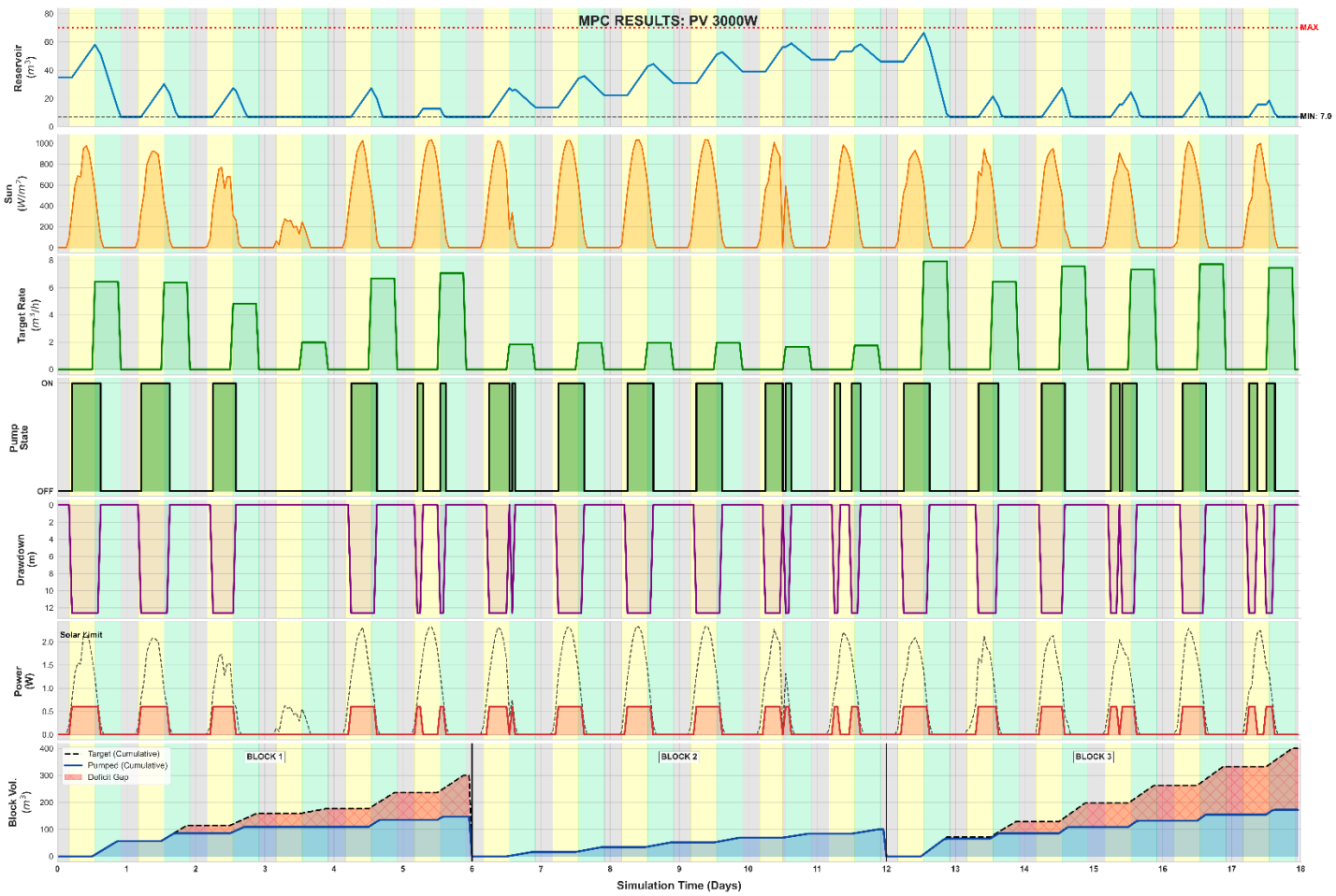

Figure 3: PV 3000W Performance in a constrained aquifer

Under these constraints, larger systems provided excessive power that drove extraction rates beyond the aquifer's recharge capacity. This led to rapid dewatering of the borehole, causing the dynamic drawdown to hit critical safety limits and triggering frequent pump lockout periods. Additionally, the higher start-up power threshold (20% of rated capacity) meant larger systems remained inactive during low-irradiance hours that the 3000 Wp system could effectively utilize.

In contrast, when simulation parameters were updated to represent an optimized, high-yield aquifer, the Oversizing Paradox was effectively mitigated. In this "relaxed" hydraulic environment, the aquifer was capable of replenishing the borehole nearly as fast as the pump extracted water.

Under these conditions, a **"more is better"** relationship was established between PV capacity and water yield. Because the high transmissivity prevented the formation of a deep cone of depression, the system was no longer limited by geological recharge rates but primarily by available solar irradiance. The 3000 Wp system achieved better water deficits even during the peak stress phase, while maintaining negligible drawdown levels. Larger systems in this scenario were able to utilize their full power potential to meet even higher demand without triggering safety cut-offs.

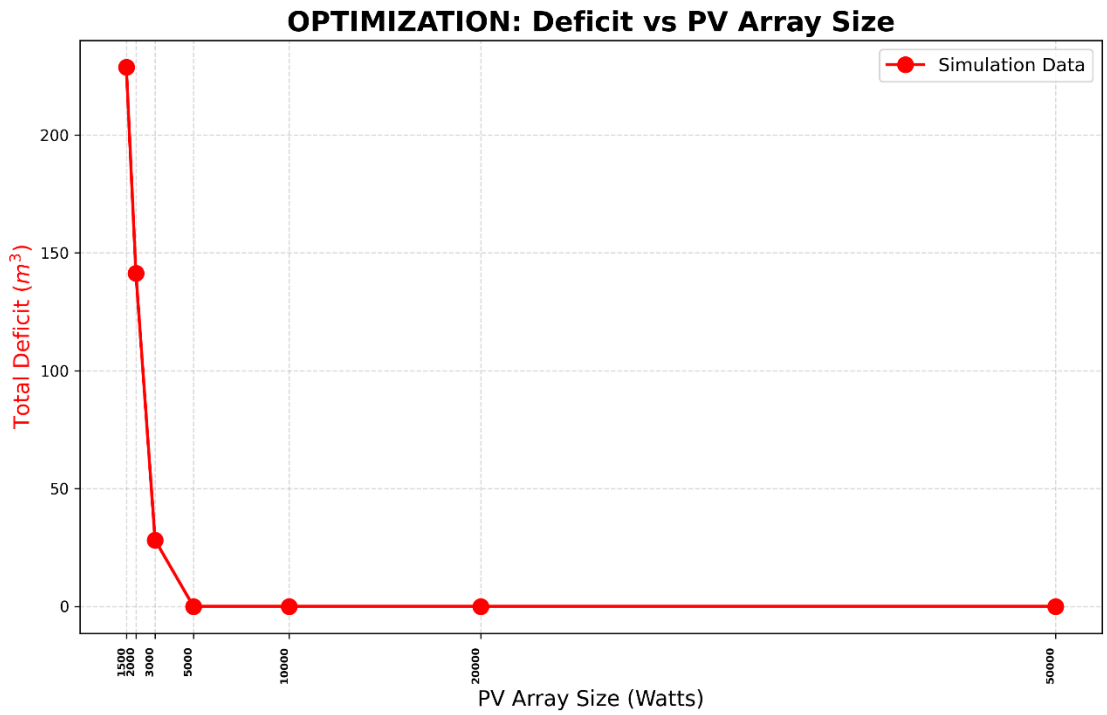

Figure 4: Deficit Analysis under Optimized Aquifer Conditions (Relaxed Case)

Figure 4: The red curve illustrates the system's performance when hydraulic constraints are relaxed. Note the monotonic decrease in deficit as PV capacity increases. The 3000 Wp system achieves a minimal deficit, while the 5000 Wp system eliminates the deficit entirely. This confirms that in high-yield aquifers, the Oversizing Paradox is resolved, and performance is limited only by solar array size.

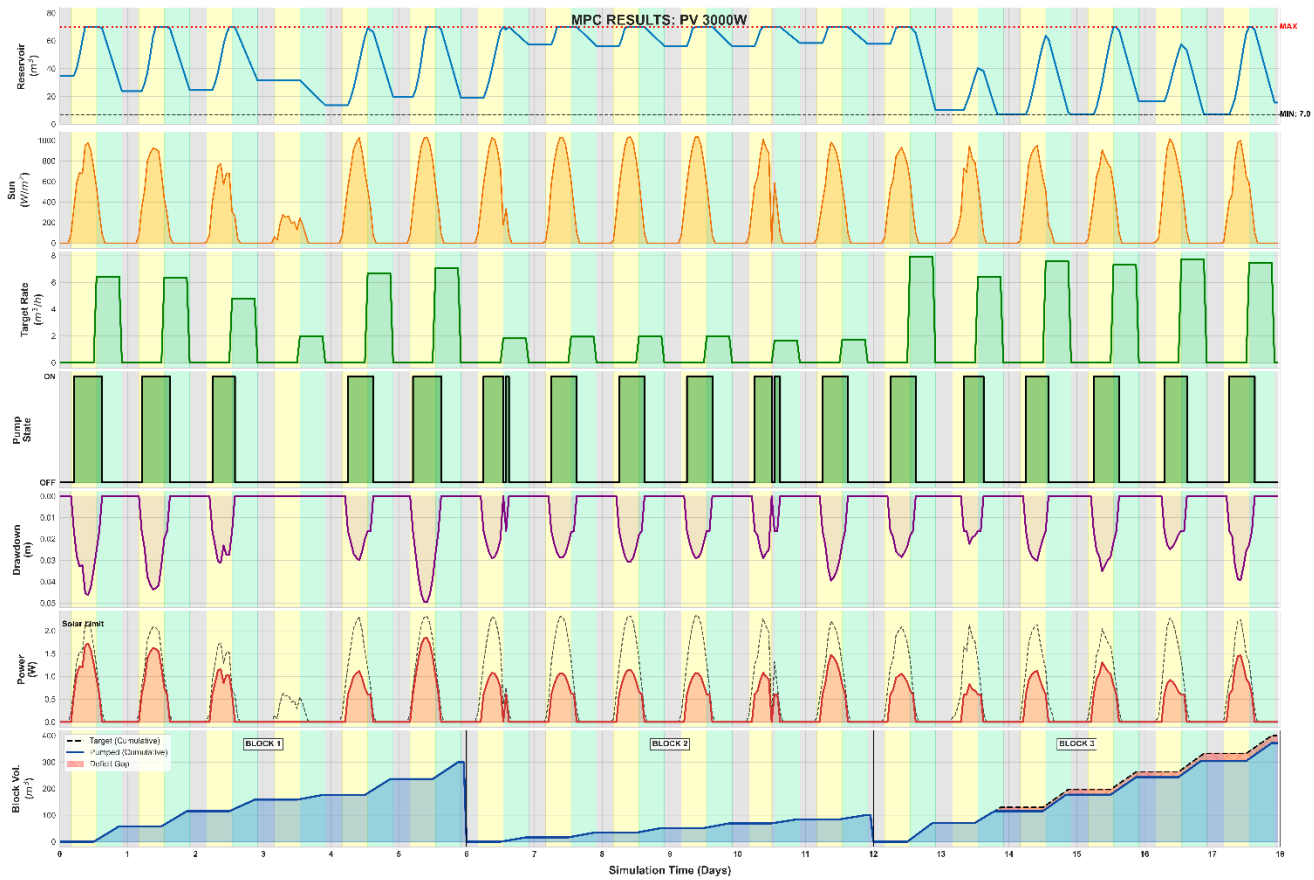

Figure 5: PV 3000W Performance in a healthy aquifer

Discussion

The simulation results underscore a fundamental shift in the design philosophy of solar pumping: the transition from a solar-limited to an **aquifer-limited** paradigm. In many hydrogeological contexts, particularly the aquifers near Nairobi, the hydraulic capacity of the borehole serves as a more rigid constraint than the peak power of the PV array. The Oversizing Paradox observed in the constrained aquifer tests highlights that simply increasing electrical power can be counterproductive, as it leads to rapid drawdown and subsequent system lockouts.

A critical observation in the constrained aquifer scenario was that increasing the **Deficit Weight** yielded no improvement in water delivery or reduction in total deficit. Under low-transmissivity conditions, the system enters a state of **hydraulic saturation**.

At this juncture, the governing constraint is no longer the controller's mathematical priority, but the physical recharge rate of the aquifer itself. Even when the controller is programmed to prioritize water volume five times more aggressively, it cannot bypass the physical hard limit where the drawdown hits the motor-pump depth. This confirms that in hydrogeologically limited

environments, the performance of a solar pumping system is decoupled from control tuning once the physical extraction limit of the borehole is reached.

The HMPC framework demonstrates resilience by actively load shaping the extraction profile. By penalizing excessive drawdown, the controller chooses to pump at lower flow rates for longer durations, effectively bypassing the hydraulic bottlenecks of the well. This behavior treats the borehole and the storage tank as a combined "virtual battery," storing potential energy during high-irradiance hours to meet irrigation demand during the evening window.

A critical finding is the role of the **20% Minimum Start-Up Threshold**. While necessary to protect the motor from low-torque stalling, this threshold creates a dead zone at sunrise and sunset. In the 3000 Wp system, 600 W of available solar power is required just to initiate flow. This suggests that future system optimizations should consider not only the peak power but also the minimum power required to start the motor-pump.

Conclusion

This study has demonstrated the application of Hierarchical Model Predictive Control to aquifer constrained solar pumping systems. By integrating the rigorous hydraulic physics defined by Meunier et al. (2023) into a real-time optimization framework, we have shown that it is possible to navigate the trade-off between maximizing water yield and preserving aquifer integrity.

The use of pvlib for real-time weather data and scipy for non-linear optimization confirms that sophisticated control logic is computationally feasible for this application. Future work should include field validation of these algorithms.

For the constrained aquifer conditions of Nairobi, a 3000 Wp system represents the optimal balance, achieving lower deficits than larger, oversized arrays. Under optimized, high-yield aquifer conditions, the system follows a "more is better" relationship, where results improve linearly with PV capacity as geological constraints are removed.

The hierarchical architecture reduced computational time by compared to monolithic optimization, proving the algorithm's suitability for edge computing deployment. Future research should recursively estimate transmissivity in real time. This will allow the HMPC to adapt dynamically to seasonal water table fluctuations and sensor noise, moving closer to a fully autonomous, hydro-aware irrigation system.

Furthermore, the transition from simulation to field deployment must address the "5Vs" of data-driven modelling (Volume, Velocity, Variety, Value, and Veracity) as identified by Bwambale et al. (2023).

References

[1] "Meunier, S., et al. (2023). Aquifer conditions, not irradiance determine the potential of photovoltaic energy.".

[2] "Bwambale, E., et al. (2023). Data-driven model predictive control for precision irrigation.".

[3] " Scattolini, R. and Colaneri, P. (2007). Hierarchical Model Predictive Control.".